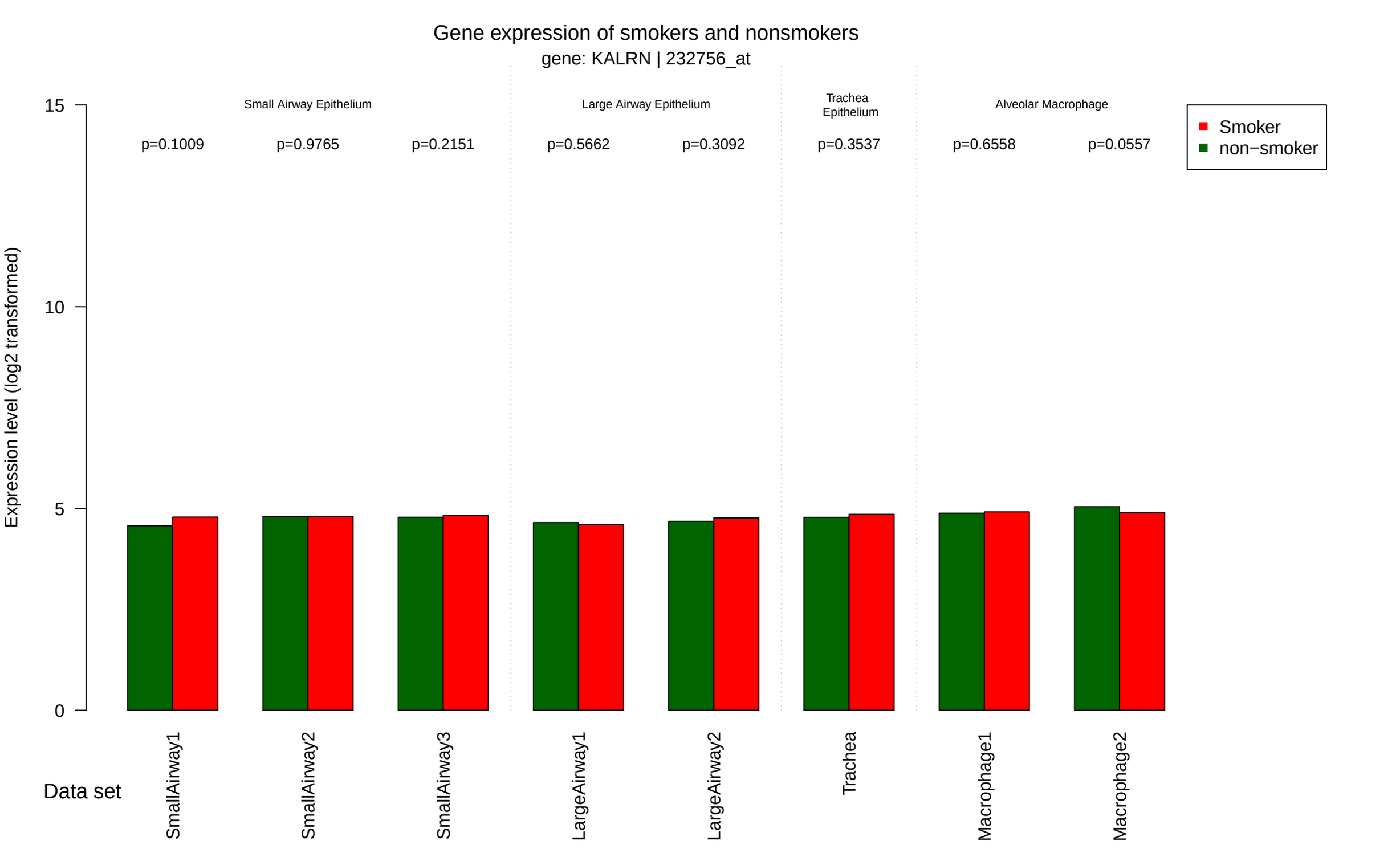
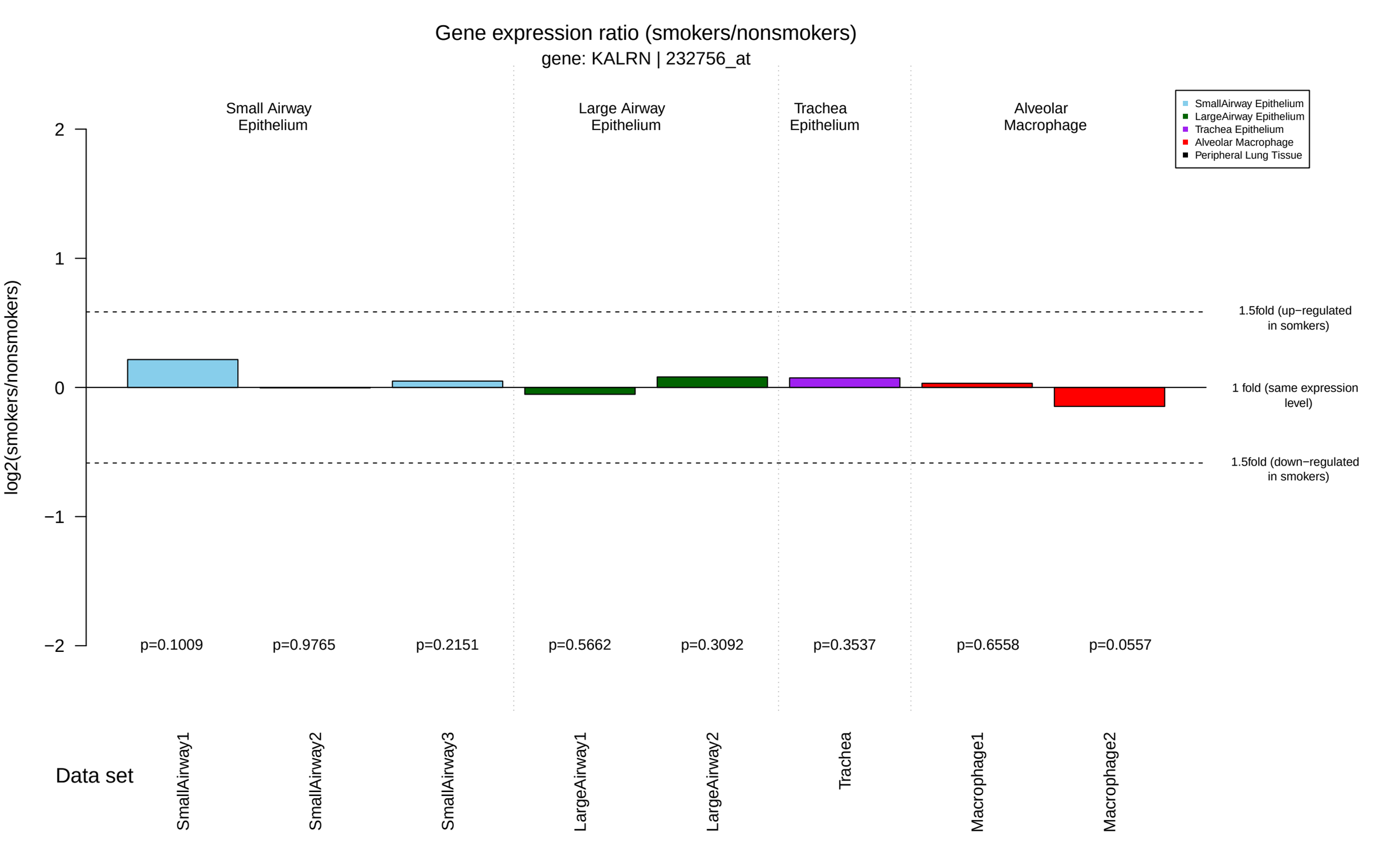
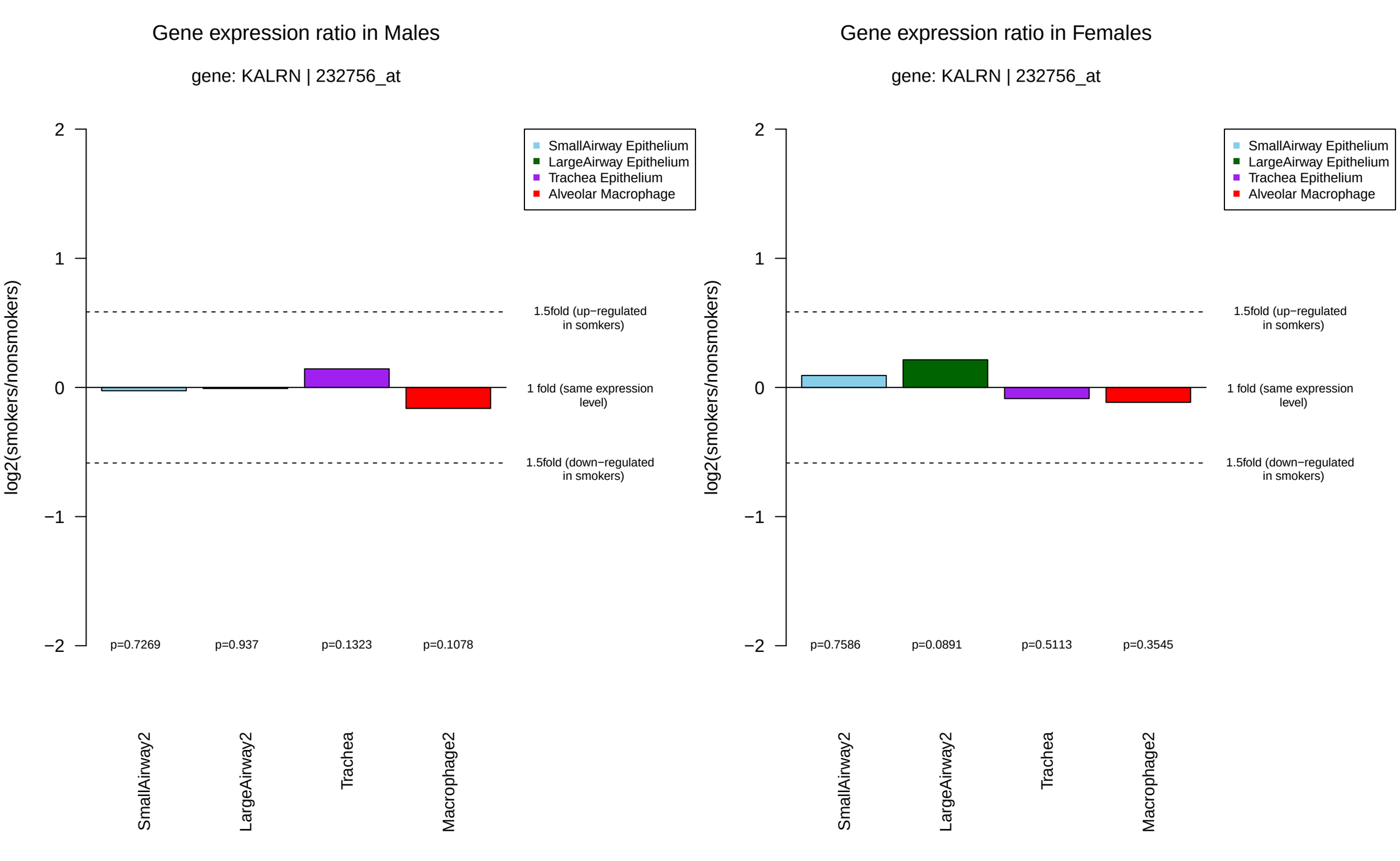
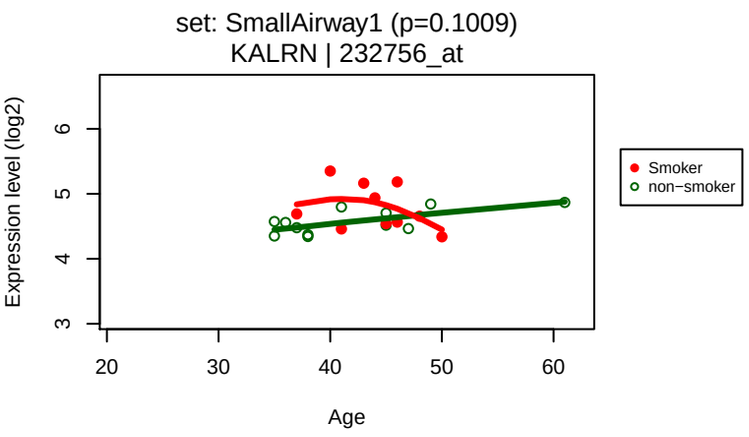
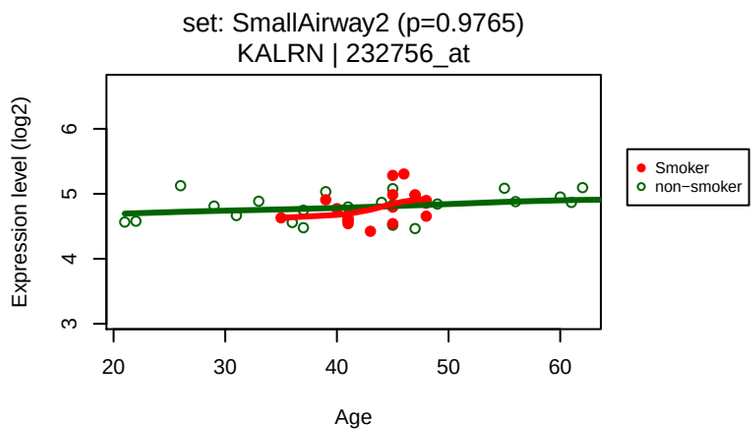
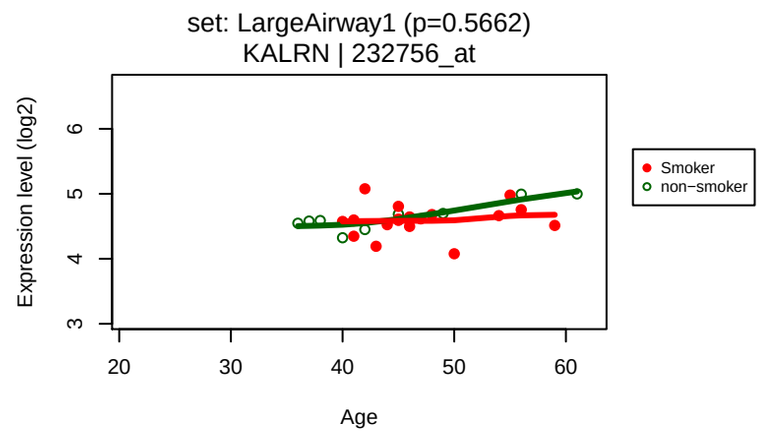

Gene expression of smokers and nonsmokers
gene: KALRN | 232756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1009
p=0.9765
p=0.2151
p=0.5662
p=0.3092
p=0.3537
p=0.6558
p=0.0557
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

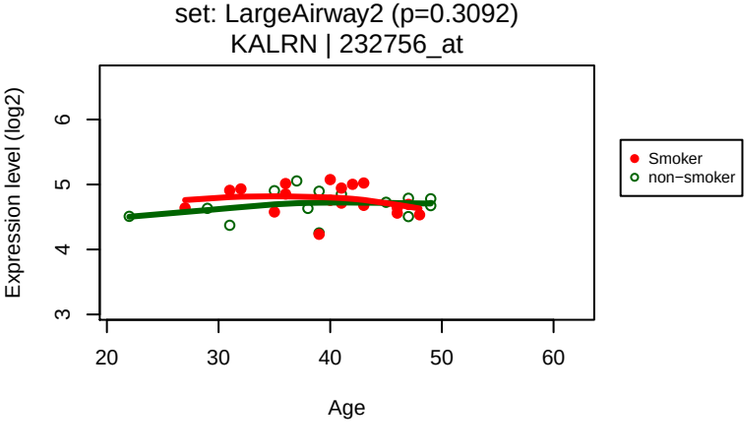

Gene expression ratio (smokers/nonsmokers)
gene: KALRN | 232756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1009
p=0.9765
p=0.2151
p=0.5662
p=0.3092
p=0.3537
p=0.6558
p=0.0557
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

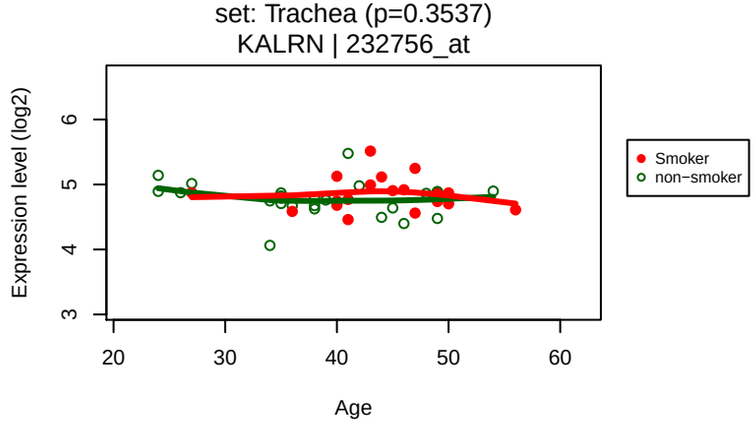

Gene expression ratio in Males
gene: KALRN | 232756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7269
p=0.937
p=0.1323
p=0.1078
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KALRN | 232756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7586
p=0.0891
p=0.5113
p=0.3545
SmallAirway2
LargeAirway2
Trachea
Macrophage2

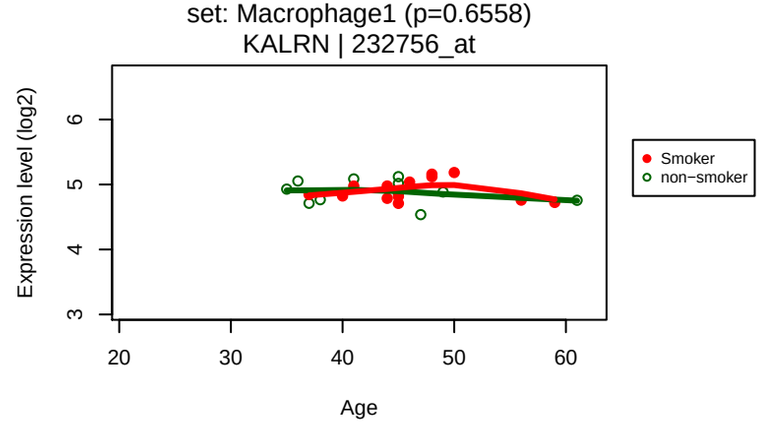

set: SmallAirway1 (p=0.1009)
KALRN | 232756_at
Expression level (log2)
Age
Smoker
non−smoker

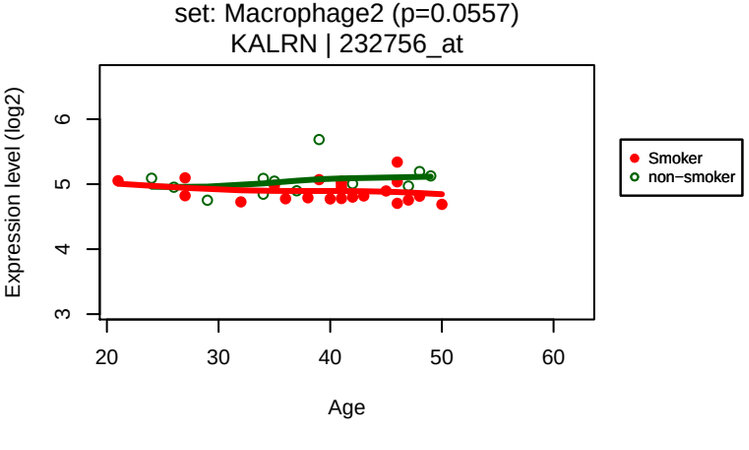

set: SmallAirway2 (p=0.9765)
KALRN | 232756_at
Expression level (log2)
Age
Smoker
non−smoker

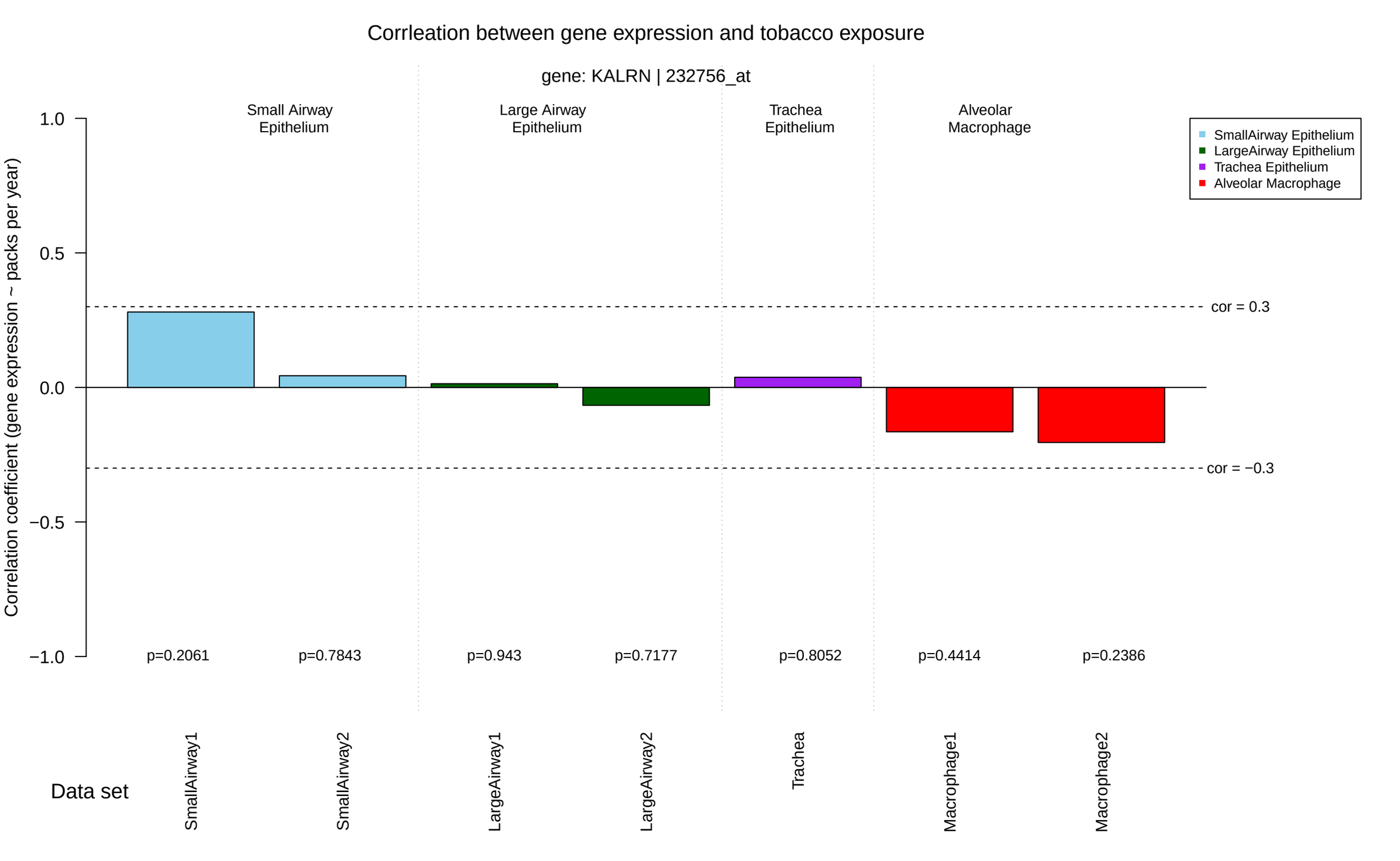

set: LargeAirway1 (p=0.5662)
KALRN | 232756_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3092)
KALRN | 232756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3537)
KALRN | 232756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6558)
KALRN | 232756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0557)
KALRN | 232756_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KALRN | 232756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2061
p=0.7843
p=0.943
p=0.7177
p=0.8052
p=0.4414
p=0.2386
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2